



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
01-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS, TNAS & TAAS]

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

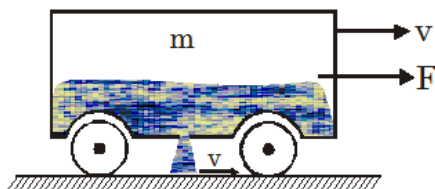
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

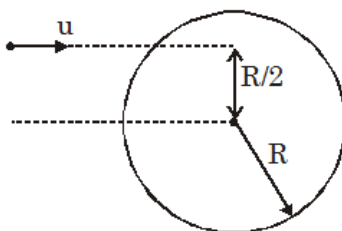
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A cart loaded with sand moves along a horizontal floor due to a constant force F coinciding in direction with the cart's velocity vector. In the process sand spills through a hole in the bottom with constant rate μ kg/sec. Find the velocity of the cart as a function of time ' t ', if at the initial moment $t = 0$ the cart with loaded sand had the mass m_0 and its velocity was equal to zero.



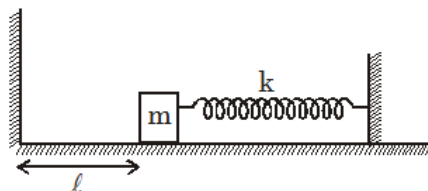
- (A) $\frac{F}{\mu} \ln \left(\frac{m_0}{m_0 - \mu t} \right)$ (B) $\frac{F}{\mu} \ln \left(\frac{m_0}{m_0 + \mu t} \right)$ (C) $\frac{F}{\mu} \ln \left(\frac{m_0 + \mu t}{m_0} \right)$ (D) $\frac{F}{\mu} \ln \left(\frac{m_0 - \mu t}{m_0} \right)$
2. A fixed non-conducting spherical shell of radius R has a charge Q spread uniformly on its surface. A particle having mass ' m ' and charge ' q ' is projected towards the shell from a large distance at a speed u . Impact parameter is $\frac{R}{2}$ as shown in figure. The particle passes the shell grazing its surface. The projection speed u of the particle is (Neglect gravity).



- (A) $\sqrt{\frac{4Qq}{3\pi\epsilon_0 mR}}$ (B) $\sqrt{\frac{Qq}{2\pi\epsilon_0 mR}}$ (C) $\sqrt{\frac{2Qq}{3\pi\epsilon_0 mR}}$ (D) $\sqrt{\frac{3Qq}{2\pi\epsilon_0 mR}}$

Space for Rough Work

3. A small block of mass m resting on a horizontal smooth floor at distance ℓ from a rigid wall. Block attached to the spring, is pushed towards right by a distance $\frac{3\ell}{2}$ and released. When block passes from its mean position an other block of mass m_1 is placed on it which sticks to it due to friction. If combined block just collides with the left wall, then :-



- (A) $m_1 = m$ (B) $m_1 = \frac{m}{4}$ (C) $m_1 = \frac{3m}{4}$ (D) $m_1 = \frac{5m}{4}$
4. A point object is placed at a distance of 15 cm from a convex lens. The image is formed on the other side of lens at a distance 30 cm from lens. When a concave lens is placed in contact with convex lens, image is shifted further by 30 cm. Calculate the focal length of concave lens :-

- (A) 10 cm (B) -10 cm (C) -60 cm (D) 60 cm

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

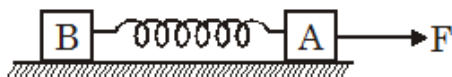
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

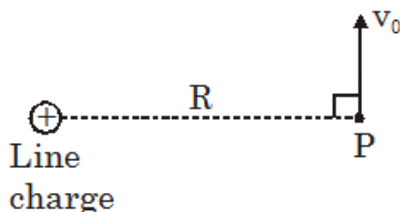
5. Two blocks A and B each of mass 'm', are connected by a massless spring of force constant k. Initially given system is at rest and spring is in its natural length. A constant force F starts acting on block A at time $t = 0$ and at time t, the extension in the spring is seen to be ℓ . Given system lies on smooth horizontal surface.



- (A) Acceleration of centre of mass of the given system (A + B + spring) is $\frac{F}{m}$.
- (B) Displacement of centre of mass of the given system (A + B + spring) in time t is $\frac{F t^2}{4m}$.
- (C) Displacement of block A in time t is $\frac{F t^2}{4m} + \frac{\ell}{2}$.
- (D) Displacement of block A in time t is $\frac{F t^2}{2m} - \frac{\ell}{2}$.

Space for Rough Work

6. An infinite line charge is perpendicular to the plane of the figure having linear charge density λ . A particle having charge $+Q$ and mass m is projected in the field of the line charge from point P. The point P is at a distance R from the line charge and velocity given to the particle is v_0 perpendicular to the radial line at P as shown in figure.



- (A) Speed of the particle when its distance from the line charge grows to ηR ($\eta > 1$), is

$$\sqrt{v_0^2 + \frac{\lambda Q}{\pi \epsilon_0 m} \ln(\eta)}$$

- (B) Speed of the particle when its distance from the line charge grows to ηR ($\eta > 1$), is

$$\sqrt{v_0^2 - \frac{\lambda Q}{\pi \epsilon_0 m} \ln(\eta)}$$

- (C) Velocity component of the particle along the radial line (joining the line charge to the particle)

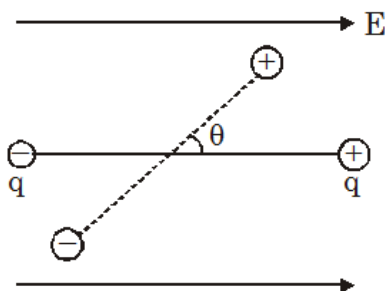
at the instant its distance is ηR , is given by $\sqrt{v_0^2 \left(1 - \frac{1}{\eta^2}\right) + \frac{\lambda Q}{\pi \epsilon_0 m} \ln(\eta)}$

- (D) Velocity component of the particle along the radial line (joining the line charge to the particle)

at the instant its distance is ηR , is given by $\sqrt{v_0^2 \left(1 + \frac{1}{\eta^2}\right) + \frac{\lambda Q}{\pi \epsilon_0 m} \ln(\eta)}$

Space for Rough Work

7. Figure shows an electric dipole formed by two particles fixed at the ends of a light rod of length ℓ . The mass of each particle is m and the charges are $-q$ and $+q$. The system is placed in such a way that the dipole axis is parallel to a uniform electric field E that exist in the region. The dipole is slightly rotated about its centre and released. Considering small angular displacement θ , which of the following options are correct.



- (A) Motion of the dipole is angular SHM with time period $2\pi\sqrt{\frac{m\ell}{2qE}}$
- (B) Motion of the dipole is angular SHM with time period $2\pi\sqrt{\frac{m\ell}{qE}}$
- (C) During SHM no net force acts on dipole.
- (D) During SHM, when dipole moment vector becomes parallel to electric field, net torque on dipole is zero.
8. Two fixed charges $-2Q$ and Q are located at the points with coordinates $(-3a, 0)$ and $(+3a, 0)$ respectively in the xy plane.
- (A) All the points in x - y plane where the electric potential due to the two charges is zero lie on a circle of radius $4a$.
- (B) All the points in x - y plane where the electric potential due to the two charges is zero lie on a circle having its centre at $(5a, 0)$
- (C) All the points in x - y plane where the electric potential due the two charges is zero lie on an ellipse.
- (D) All the points in x - y plane where the electric potential due to the two charges is zero lie on an ellipse having its centre at $(5a, 0)$

Space for Rough Work

9. A horizontal spring mass system of mass M executes oscillatory motion of amplitude a_0 and time period T_0 . When the mass M is passing through its equilibrium position another mass m is placed on it such that both move together. If a and T be the new amplitude and time period respectively then,

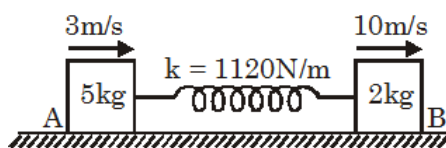
(A) $a = \sqrt{\frac{M}{M+m}} a_0$

(B) $a = \sqrt{\frac{M+m}{M}} a_0$

(C) $T = \sqrt{\frac{M}{M+m}} T_0$

(D) $T = \sqrt{\frac{M+m}{M}} T_0$

10. Two blocks A(5kg) and B(2kg) attached to the ends of a spring of spring constant 1120 N/m are placed on a smooth horizontal plane with the spring undeformed. Simultaneously velocities of 3m/s and 10 m/s along the line of the spring in the same direction are imparted to A and B at $t = 0$ then:



- (A) When the extension of the spring is maximum the velocities of A and B are zero.
- (B) The maximum extension of the spring is 25 cm.
- (C) The first maximum compression occurs at $t = \frac{3\pi}{56}$ second.
- (D) Maximum extension and maximum compression occur alternately.

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

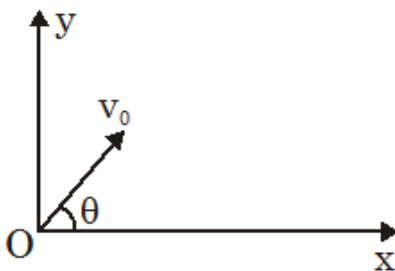
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A projectile is thrown from a point 'O' on the ground at an angle θ from horizontal with initial velocity v_0 at $t = 0$. At the highest point of its trajectory, the projectile splits into two parts having mass in the ratio 1 : 2. After splitting, smaller part falls vertically downwards with initial velocity zero. The other part 't' second after splitting falls on the ground at a distance x meter from the point of projection. (Consider $v_0 = 20$ m/s, $\theta = 37^\circ$ & acceleration due to gravity $g = 10$ m/s²)



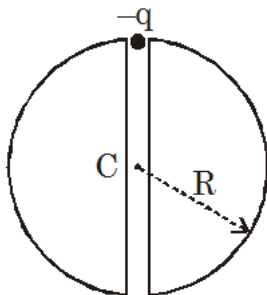
1. The value of t is _____
2. The value of x is _____

Space for Rough Work

Paragraph for Questions 3 and 4

In a uniformly charged fixed nonconducting sphere, a very thin tunnel has been made along the diameter as shown in figure. A charge particle $-q$ having mass m is released from rest at $t = 0$ at one end of the tunnel. Total charge Q of sphere is uniformly distributed in its volume. Charge particle reaches at the

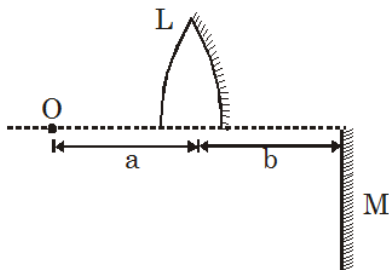
centre of solid sphere first time at $t = \pi \sqrt{\frac{\alpha \cdot \pi \epsilon_0 m R^3}{Q \cdot q}}$ with speed $v = \sqrt{\frac{Qq}{N \cdot \pi \epsilon_0 m R}}$.



3. The value of N is _____
4. The value of α is _____

Paragraph for Questions 5 and 6

In figure L is half part of an equiconvex lens of refractive index 1.5 whose surfaces have radius of curvature 40 cm and its right surface is silvered as shown in figure. Normal to its principal axis, a plane mirror M is placed on right of lens. Distance between lens L and mirror M is b . A small object O is placed on left of the lens. Images found by lens and mirror coincide with each other and transverse length of final image formed by the lens is twice that of image formed by the mirror.



5. The value of a (in cm) is _____
6. The value of b (in cm) is _____

Space for Rough Work

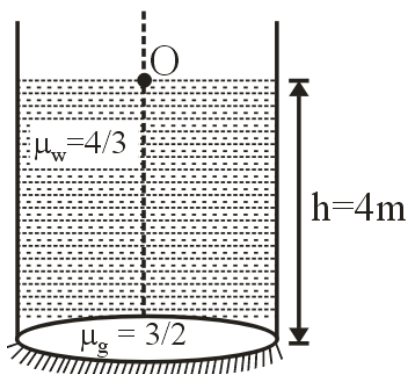
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. In a compound microscope the objective and eye piece have focal lengths of 0.95 cm and 5 cm respectively, and are kept at a distance of 20 cm. The last image is formed at a distance of 25 cm from eye piece. Calculate the magnitude of total magnification of the compound microscope.
8. Bottom of a glass beaker is made of a thin equi-convex lens having bottom side silvered as shown in figure. Water is filled in the beaker up to a height 4m. The image of a point object O, floating at middle point of beaker at the surface of water coincides with it. Find out the radius (in cm) of curvature of the lens. Given that refractive index of glass is $\frac{3}{2}$ and that of water is $\frac{4}{3}$.



9. A ring of radius R that lies in the yz plane carries a positive charge Q uniformly distributed over its circumference. A particle of mass m that carries a negative charge of magnitude q is at the center of the ring. When the charges Q and q are $5\text{ }\mu\text{C}$ and $-5\text{ }\mu\text{C}$, respectively, and the radius of the ring is 8.0 cm, the mass m oscillates about its equilibrium position with an angular frequency of 16 rad/s. Find the angular frequency (in rad/s) of oscillation of the mass if the radius of the ring is made four times to 32 cm and all other parameters remain unchanged. (Round off to nearest integer).

Space for Rough Work

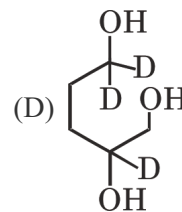
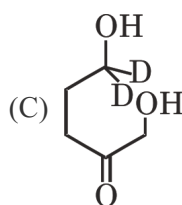
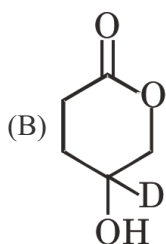
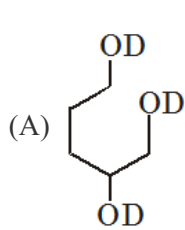
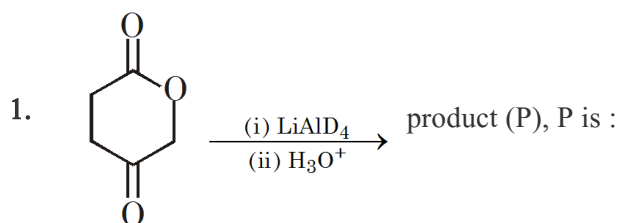
PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

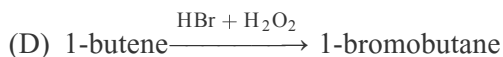
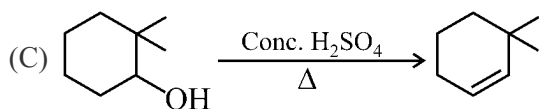
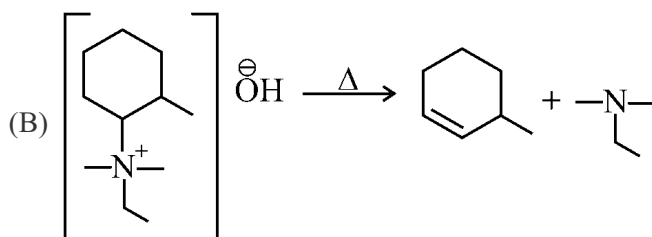
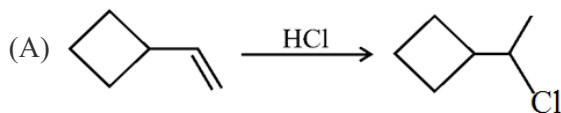
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases



2. Reaction correctly matched with its major product is :



Space for Rough Work

3. An ideal binary solution contains 4 mol of A and 6 mol of B. Its vapour pressure is found to be 460 mmHg. To this solution, when 10 mol of B is added, its vapour pressure becomes 480 mmHg. The vapour pressures of pure A and pure B, respectively, are
- (A) 300 mmHg, 600 mmHg (B) 400 mmHg and 500 mmHg
(C) 300 mmHg, 500 mmHg (D) 400 mmHg and 600 mmHg
4. Which of the following does not obey Sidgwick effective atomic number rule -
- (A) $\text{Co}_2(\text{CO})_8$ (B) $\text{Fe}(\text{CO})_5$ (C) $[\text{V}(\text{CO})_6]^-$ (D) $[\text{Mn}(\text{CO})_6]$
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

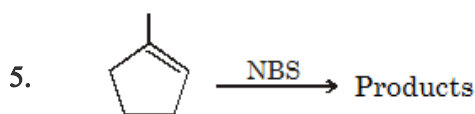
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

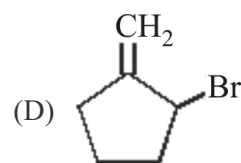
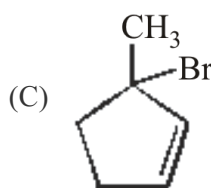
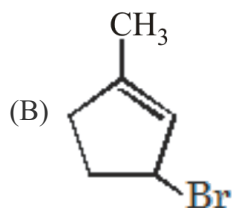
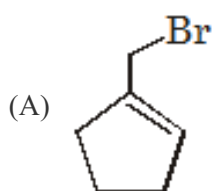
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

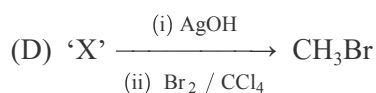
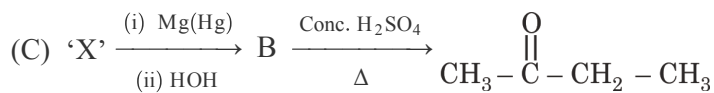
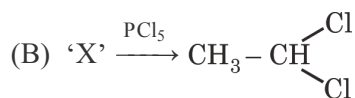
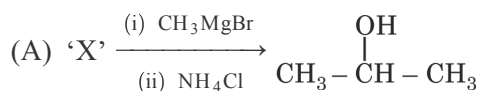


Possible product(s) is/are :



Space for Rough Work

6. Out of the given options reaction(s) in which the substrate 'X' can be acetaldehyde to give the final product :



7. Which of the following statement(s) is/are correct ?

- (A) In fluorite structure, tetrahedral voids are occupied by anions.
 (B) In zinc-blende structure, half of tetrahedral voids are occupied by cations.
 (C) In rock-salt structure, cations occupy tetrahedral voids.
 (D) The general formula of an ionic compound crystallizing in antifluorite structure is A_2B .

Space for Rough Work

8. Which of the following statements are not correct ?
- (A) For a reaction $(1/2)A \rightarrow 2B$, the rate of disappearance of A is $-2 d[A]/dt$
- (B) For a reaction $(1/2)A_2 \rightarrow 2B$, the rate of disappearance of A_2 is related to the rate of appearance of B through an expression - $d[A_2]/dt = 4 d[B]/dt$.
- (C) Reactions with small activation energies show a lesser increase in rate constant with rise in temperature as compared to that having high activation energy.
- (D) If the half-life for a given reaction having single reactant is doubled as the initial concentration of a reactant was doubled then order of the reaction would be zero.
9. $M^{2+} + NaH_2EDTA \rightarrow [M(EDTA)]^{2-} + 2H^+ + 2Na^+$
Which of the following metal ion has least tendency for the above reaction.
- (A) Cs^+ (B) Mg^{2+} (C) Ca^{2+} (D) Pb^{2+}
10. Select similarity between : $[Co(NH_3)_4(H_2O)Cl]Cl_2$ and $[Co(NH_3)_4Cl_2]Cl \cdot H_2O$
- (A) Oxidation number and coordination number of metal
- (B) Cl^- and H_2O are the ligands
- (C) Amount of precipitate with $AgNO_3$ solution (by equimolar 1 litre solution of both)
- (D) Number of Co-N linkages

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

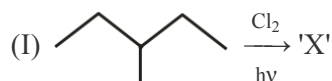
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider the reactions given below and answer the following questions



1. Find out total number of monochlorinated product 'X'
2. If the mixture of dichlorinated products (Y) is subjected to fractional distillation, how many fractions would be obtained?

Paragraph for Questions 3 and 4

For the zero-order reaction :



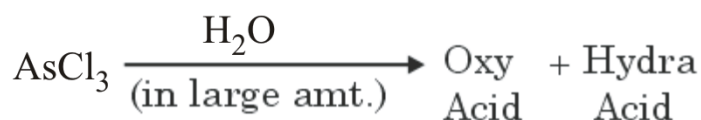
At $t = 0$, partial pressure of A = 4000 mm Hg

Rate constant of reaction = 5 mm Hg /sec

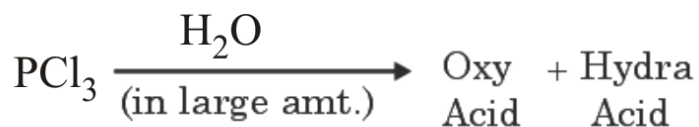
3. Find the total final pressure of gas mixture at $t = 250$ second (in mm Hg)
4. Find the partial pressure of 'B' at $t = 100$ second (in mm Hg)

Space for Rough Work

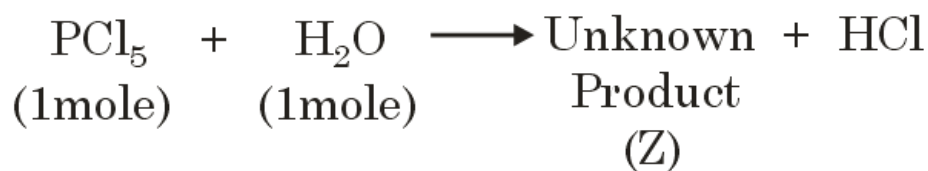
Paragraph for Questions 5 and 6



↓
Basicity(X)



↓
Basicity(Y)



5. The value of $\left(\frac{X}{Y}\right)$ is.....
6. The ratio of number of π bond(s) to number of σ bond(s) in 'Z' is

Space for Rough Work

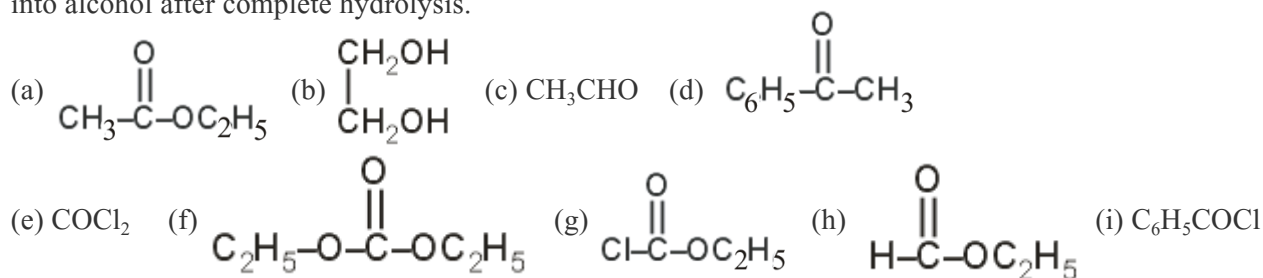
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

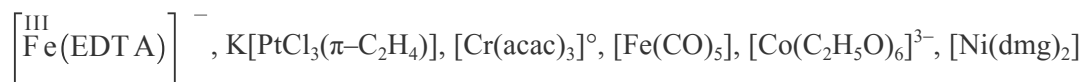
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. How many compounds out of following use two or more than two moles of Grignard reagent to convert into alcohol after complete hydrolysis.



8. For the reaction $2\text{NO} + \text{H}_2 \rightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$ the value of $-\text{dp}/\text{dt}$ was found to be 1.50 Torr s^{-1} for a pressure of 304 Torr of NO and $0.375 \text{ Torr s}^{-1}$ for a pressure of 152 Torr of NO, the pressure of H_2 being constant. On the other hand, when the pressure of NO was kept constant, $-\text{dp}/\text{dt}$ was 1.60 Torr s^{-1} for a hydrogen pressure of 294 Torr and 0.8 Torr s^{-1} for a hydrogen pressure of 147 Torr. Determine the overall order of the reaction.
9. Select complex in which metal-carbon linkage(s) is / are present :



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $y = \frac{ax+b}{cx+d}$, then value of $\frac{y_2^2}{y_1 y_3}$, wherever exists, is (where y_n denotes n^{th} order derivative of differentiable function y and a, b, c, d are constants) -

(A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $\frac{4}{5}$ (D) $\frac{5}{4}$

2. If $f(x) = \begin{vmatrix} (x-1)^2 & (x-2)^3 & (x+1) \\ (x-1)^3 & x^2 & (x+1)^2 \\ x+2 & x+1 & x^5 \end{vmatrix}$ then coefficient of x in $f(x)$ is -

(A) -14 (B) -16 (C) 0 (D) 25

Space for Rough Work

3. Let $f(x)$ be a function satisfying the condition $f(|x - 3|) + 2f(|x + 3|) = x^2 + 2x \forall x \in \mathbb{R}$, then which of the following is not correct
- (A) minimum value of $f(x)$ is -3 (B) $f(x)$ is one-one function
(C) equation $|f(x)| = 2$ has four distinct solutions (D) $|f(x)|$ is non-differentiable at two points
4. For three invertible functions, $f(x) = \frac{x-3}{x+3}$, $g(x) = 2x$ and $h(x) = 2^{\frac{x+3}{x+2}}$, value of $f^{-1}(g^{-1}(h^{-1}(4)))$ is -
- (A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) 1 (D) -1
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Given $\lim_{x \rightarrow \alpha} \frac{\sin([x^3] - [x]^3)}{(x - \alpha)^3}$ exists finitely, then the value of α can be (where $\alpha \in I$ & $[.]$ denotes greatest integer function) -
- (A) 0 (B) 1 (C) -1 (D) 2
6. Let $f : \mathbb{R} \rightarrow A$ be an invertible function defined by $f(x) = \log_2(2^{x^3+x} + 4)$. If $g(x)$ be its inverse function, then -
- (A) $g(3) = 2$ (B) $g'(3) = \frac{1}{2}$ (C) $\lim_{x \rightarrow 3} (g(x))^{\frac{\ln 4}{x-3}} = 2$ (D) set A is $(2, \infty)$
7. If the function $f(x) = -4e^{\frac{1-x}{2}} + 1 + x + \frac{x^2}{2} + \frac{x^3}{3}$ and $g(x) = f^{-1}(x)$, then -
- (A) $g'\left(-\frac{7}{6}\right) = \frac{1}{5}$ (B) $g'\left(\frac{7}{6}\right) = \frac{1}{5}$ (C) $g(1) = -\frac{7}{6}$ (D) $g\left(-\frac{7}{6}\right) = 1$

Space for Rough Work

8. For which of the following values of 'm' the line $y = mx$ cuts the graph of $y = \sin^{-1}(\sin x)$ in exactly 5 points -
- (A) $\frac{1}{5}$ (B) $-\frac{1}{4}$ (C) $-\frac{1}{8}$ (D) $-\frac{1}{9}$
9. Let $f(x) = x - 1$ & $g(x) = \frac{1}{x}$, then the set of points, where $[(g \circ f \circ g)(x)]$ is discontinuous is ($[.]$ denotes greatest integer function) -
- (A) $\{0\}$ (B) $\left\{ \frac{n}{n+1} \right\}, n \in I - \{-1\}$
- (C) $\left\{ \frac{n}{n+1} \right\} \cup \{1\}, n \in I - \{-1\}$ (D) $\{0, 1\}$
10. If $\sec^{-1} \left(\sqrt{1+x^2} \right) + \operatorname{cosec}^{-1} \left(\frac{\sqrt{1+y^2}}{y} \right) + \cot^{-1} \left(\frac{1}{z} \right) = \pi$ ($x, y, z > 0$) then $x + y + z$ is always -
- (A) equal to xyz (B) greater than or equal to $3(xyz)^{1/3}$
- (C) greater than or equal to $4(xyz)^{1/3}$ (D) greater than or equal to $5(xyz)^{1/3}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Given $g(x) = \sin \frac{1}{x}$, $h(x) = x^2 + 2x + (\lambda + 1)$ and $u(x) = \frac{1}{x} + \cos \frac{1}{x^2}$

Let $f(x) = \lim_{n \rightarrow \infty} \frac{x^{2n+1}g(x) + h(x)}{x^{2n} + 3x \cdot u(x)}$

1. If $\lim_{x \rightarrow 0} f(x) = 2$, then the value of λ is -
2. $\lim_{x \rightarrow \infty} f(x)$ is equal to -

Paragraph for Questions 3 and 4

Let $f : \mathbb{R} \rightarrow [0, \infty)$ be a derivable function which satisfies $f(x^2 + y^2) = f(2xy) + f(x^2 - y^2) \forall x, y \in \mathbb{R}$. Also $f'(1) = 2$.

3. $f(3) + f(4)$ is equal to -
4. Number of solutions of $f(\cos \theta) = f(\sin \theta)$ in $[0, 2\pi]$ -

Space for Rough Work

Paragraph for Questions 5 and 6

Consider $f(x) = x^4 - 3x^3 + 2$ and $h(x) = \frac{f(x)}{g(x)}$ be a rational function.

- (i) $\lim_{x \rightarrow \infty} h(x) = \text{infinite}$
 - (ii) $\lim_{x \rightarrow 1} h(x) = \frac{5}{2}$
 - (iii) $h(x)$ is continuous $\forall x \in \mathbb{R} - \{1, 2\}$
 - (iv) $g(x)$ does not have any repeated factors.
5. Number of points where $P(x) = \text{sgn}(g(x))$ is non differentiable in $\mathbb{R}$ is (where $\text{sgn}(x)$ denotes signum function of x)
6. Number of solutions of equation $|g(|x|)| = \frac{1}{2}$ is -

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. If $[x]^3 + [y]^3 + [z]^3 - 3[x][y][z] = 4$ where $x, y, z > 0$ then maximum possible value of $([x] + [y])^{[z]}$ is (where $[.]$ represents greatest integer function)
8. If a particle starts moving from $(0, 0)$ and rate of change of its abscissa is 8 cm/s on the curve $y = \sqrt{x}$, then rate of change of area of triangle formed by co-ordinate axes & tangent at a point reached after 2 sec of start is
9. If $y = \left(\ln \left(x + \sqrt{x^2 + 1} \right) \right)^2$ and $(x^2 + 1) \frac{d^3y}{dx^3} + (px + q) \frac{d^2y}{dx^2} + r \frac{dy}{dx} = sy$, then value of $(p + q + r + s)$ is
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work